

Healthcare Readmission Prediction

Applied ML & Intelligent Decision Systems Suite (2025) · Notebook 1 of 4

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0.6613 ROC-AUC Best model score	0.89 Precision Not readmitted class	101,766 Records Total encounters	~11% Imbalance 30-day readmission rate
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1. Project Overview

This notebook tackles a high-impact clinical problem: predicting whether a diabetic patient will be readmitted to hospital within 30 days of discharge. Early identification of high-risk patients allows healthcare providers to schedule follow-up care, adjust treatment plans, and reduce the human and financial cost of preventable readmissions — a problem estimated to cost the US healthcare system over \$26 billion annually.

The dataset used is the UCI Diabetes 130-US Hospitals dataset, comprising 101,766 inpatient encounters from 130 US hospitals between 1999 and 2008. Each record contains 50 features including patient demographics, diagnosis codes, lab results, medications, and prior visit history. The target variable is whether the patient was readmitted within 30 days of discharge — a binary outcome with significant class imbalance, as only approximately 11% of encounters result in early readmission.

2. Methodology

2.1 Data Preprocessing

The dataset presented several real-world data quality challenges that required careful handling before modelling:

- Missing values encoded as '?' were replaced with NaN. Rows with null values in the three primary diagnosis columns (diag_1, diag_2, diag_3) were dropped as these are clinically essential features that cannot be reliably imputed.
- The weight column, which contained '?' for over 96% of records, was dropped entirely rather than imputed to avoid introducing noise.
- Remaining missing values in categorical columns were filled with the label 'Unknown' to preserve row count.
- All categorical columns were encoded numerically using LabelEncoder, converting string labels to integer representations compatible with tree-based models.
- The target was binarised: encounters with readmitted = '<30' were labelled 1 (positive class); all others were labelled 0.
- SMOTE was applied exclusively to the training set after the train-test split, generating synthetic minority samples to balance the class distribution without contaminating the test evaluation.

2.2 Models Trained

Three classifiers were trained on the SMOTE-resampled training data and evaluated on the original imbalanced held-out test set:

Model	ROC-AUC	Notes
Logistic Regression	0.6069	Linear baseline
Random Forest	0.6613	Best model — selected
XGBoost	0.6177	Gradient boosting

Random Forest achieved the highest ROC-AUC of 0.6613 and was selected as the production model. While these scores are moderate compared to datasets with cleaner signal, they are consistent with published benchmarks on this specific dataset — clinical readmission prediction is an inherently difficult problem due to the many unobserved factors (patient behaviour post-discharge, home environment, medication adherence) that the dataset cannot capture.

3. Key Findings

3.1 Understanding the ROC-AUC Score

A ROC-AUC of 0.6613 on this dataset should be interpreted carefully. The score of 0.50 represents a random classifier and 1.00 represents a perfect one. Published academic work on the UCI Diabetes 130-US dataset consistently reports scores in the 0.62–0.70 range, reflecting the fundamental difficulty of predicting readmission from structured hospital records alone. The model performs meaningfully better than chance and provides genuine signal for clinical decision support — it is not a failure of the pipeline, but a reflection of the problem's complexity.

3.2 Class Imbalance and SMOTE

With only ~11% of encounters resulting in 30-day readmission, the dataset exhibits meaningful class imbalance. The classification report for Random Forest shows high precision (0.89) on the majority class but near-zero recall on the minority (readmitted) class before threshold tuning. This is the classic precision-recall trade-off under imbalance: the model is conservative about predicting readmission. SMOTE addressed the training distribution, but the test set was intentionally left imbalanced to reflect the true deployment environment. In a clinical deployment, the classification threshold would be lowered to prioritise recall — catching more at-risk patients — at the cost of some precision.

3.3 Feature Insights via SHAP

SHAP (SHapley Additive exPlanations) analysis on the Random Forest model revealed that `number_inpatient` (number of prior inpatient visits), `number_diagnoses`, and `number_emergency` were the strongest predictors of 30-day readmission. This aligns with clinical intuition: patients with frequent prior hospitalisations and complex comorbidities are at higher risk of early readmission. Medication features such as `insulin` and `metformin` showed moderate importance, reflecting their role in diabetes management quality during the index visit.

4. Technical Stack

Category	Tools / Libraries
Language	Python 3.10+
ML Framework	Scikit-learn, XGBoost
Imbalance Handling	imbalanced-learn (SMOTE)
Explainability	SHAP (TreeExplainer)

Evaluation	ROC-AUC, Classification Report, Confusion Matrix, ROC Curve
Visualisation	Matplotlib, Seaborn
Model Persistence	joblib (.pkl export)
Dataset	UCI Diabetes 130-US Hospitals (101,766 records, 50 features)
